

Thinking Styles and Look-Around Behavior: A Task-Dependent Relationship.

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- Background & Motivation -

Dual process theory (Wason & Evans, 1975)

- We have two systems for cognitive information processing:
 1. Intuitive, rapid and automatic system.
 2. Rational, slow and effortful system.

Thinking style (Epstein et al., 1996)

- Individual traits of ‘how they think’. Based on the dual process theory, it’s organized upon four components (Newton et al., 2024):
 - **Preference for Intuitive/Effortful Thinking (PIT, PET):** Preference for the respective systems.
 - **Actively Open-minded Thinking (AOT):** The tendency to weigh new information.
 - **Close-minded Thinking (CMT):** The tendency of dichotomous thinking.

Information gathering and decision-making in navigation

- People gather information to cope with various uncertainties during navigation (e.g., Keller et al., 2021; Brunyé et al., 2017).
- Captured by behavioral metrics such as look-around (Nanahara & Lee, 2025).

Potential relations between navigation and thinking styles

- Thinking styles affect people’s information gathering in decision-making in various contexts.
- Spatial information gathering is an important part of navigation because it supports memorizing and recognizing the surrounding environment.
- **Aim of the study:** To investigate relationships between thinking styles and information gathering under several navigation contexts.

- Methods: An Experiment in a Desktop VR Environment -

Participants: 149 adults ($M_{age} = 39.94$; $SD = 9.78$; $N_{male} = 86$)

Tasks and flow of the experiment (Fig. 1, 2, 3)

- **Free-exploration task:** Participants freely walked in a building for 2 minutes.
- **Object collection task:** They were asked to seek 7 objects that appeared one-by-one, within 12 minutes.
- **3D Judgement of relative direction (JRD) task:** On the completion of the object collection task, they indicated the direction of a given object from the location of another object. Repeated for three pairs of objects.
- **Navigation task:** After the 3D-JRD task, participants were asked to head toward a target object from another object.
- **Questionnaire survey:**
 - 4-Components Thinking Style Questionnaire (Newton et al., 2024)
 - Santa Barbara Sense of Direction Scale (Hegarty et al., 2002)
 - Demographic information (age, sex).

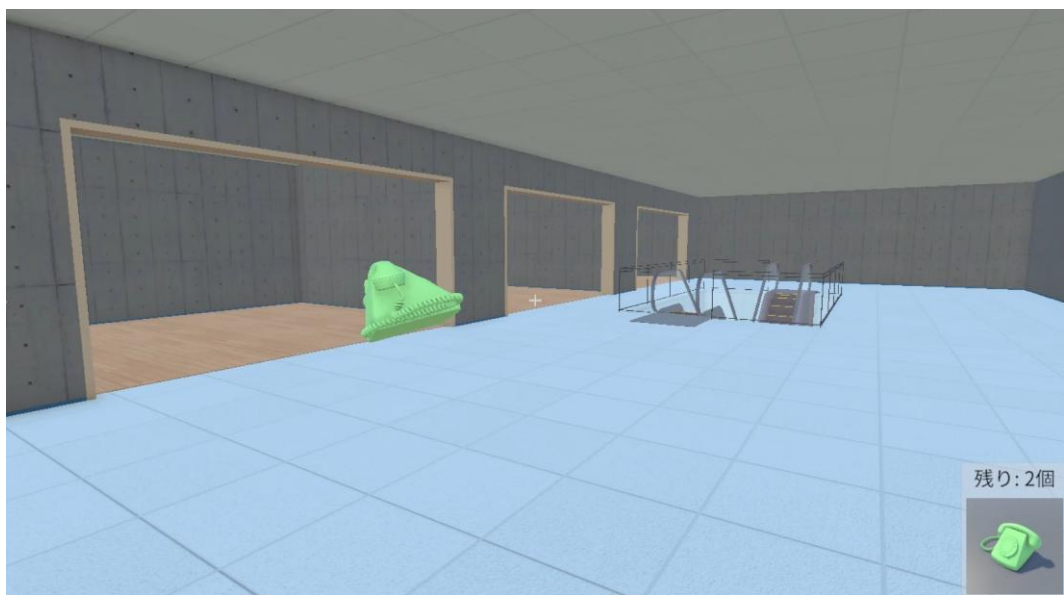


Fig. 1 Example of view during the experiment. See video from the QR code at the bottom right.

During the three tasks except for the 3D-JRD task, participants’ location (x, y, z) and camera orientation (x, y, z) were recorded in 30 fps.

Quantification of look-around behavior (LA[°])

- Accumulated angular change in camera direction during stops.
- LA is defined as the sum of the above angular changes over a trial.

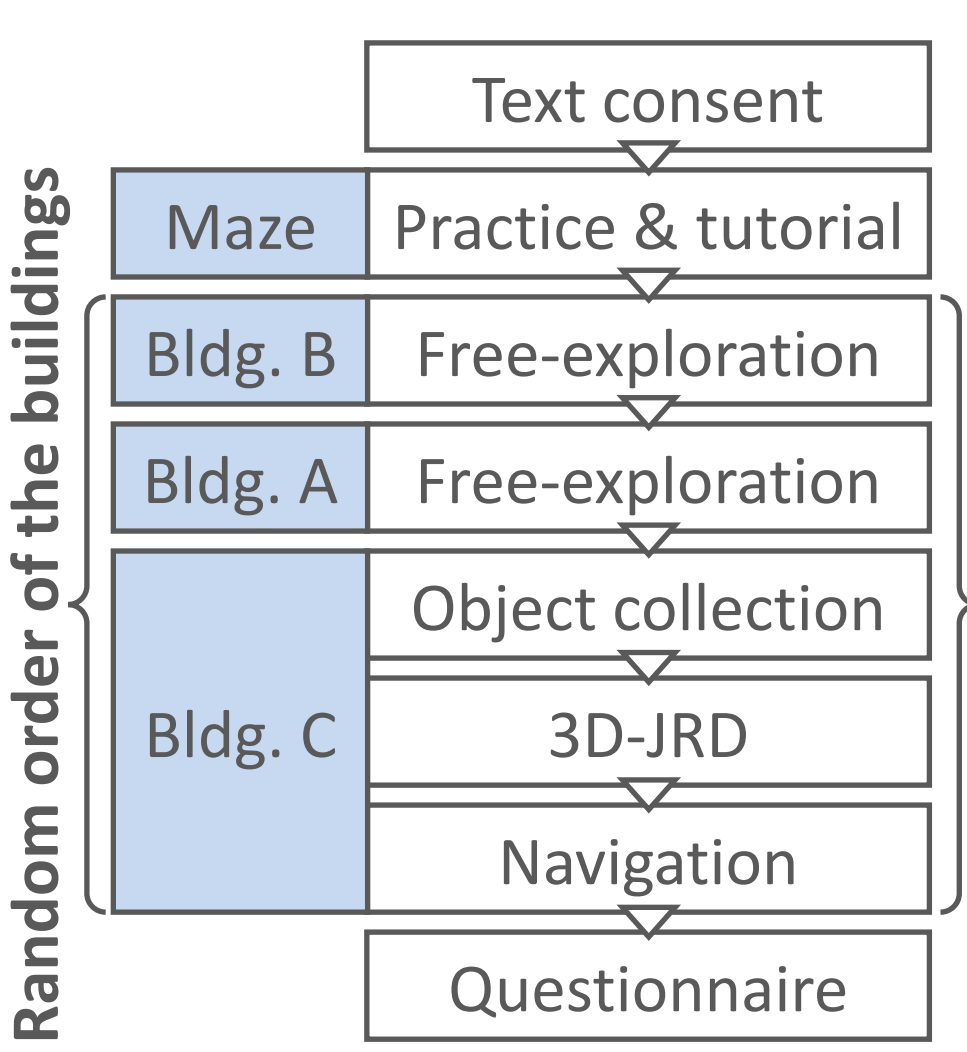


Fig. 2 Flow of the experiment

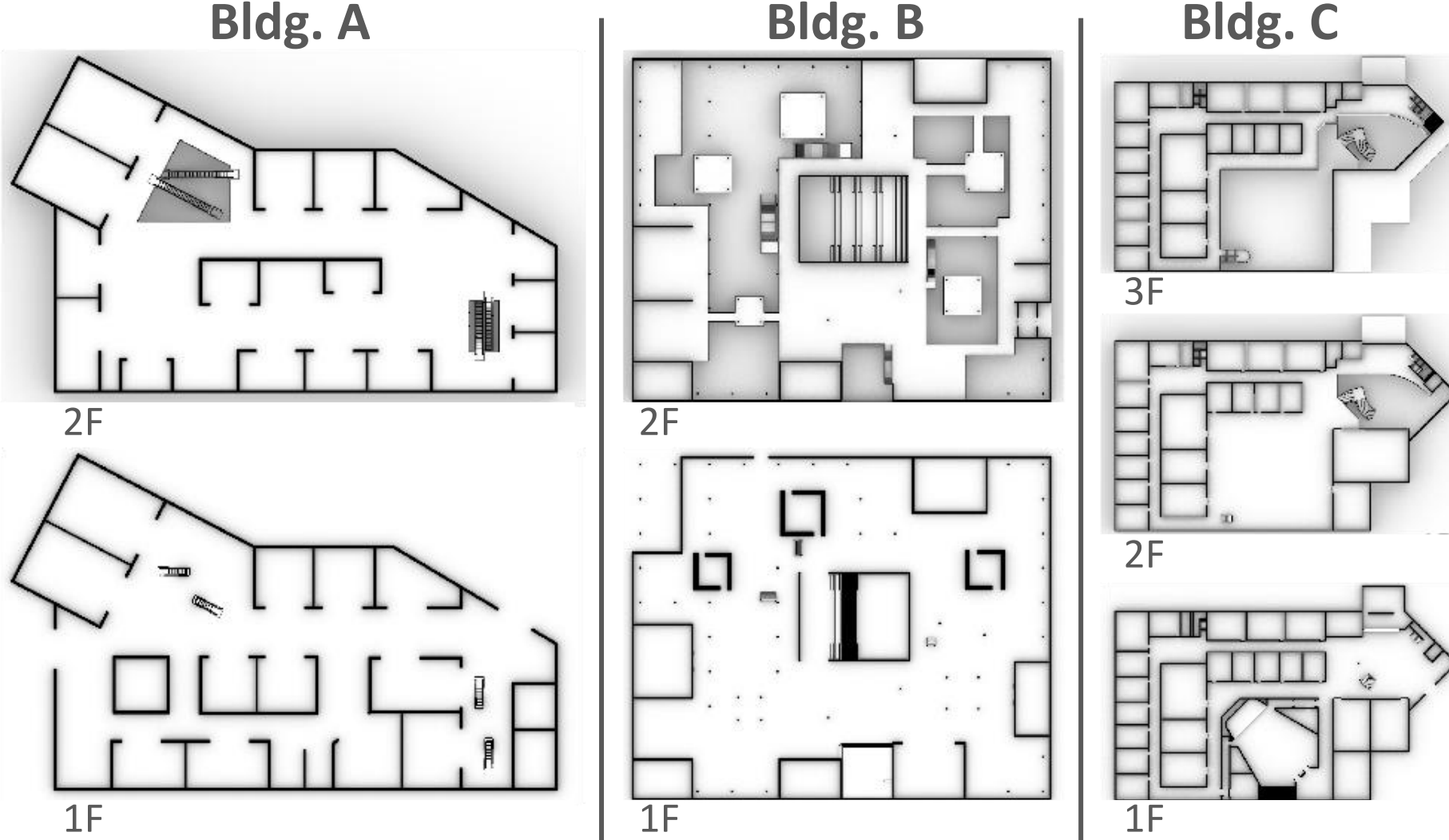


Fig. 3 Plans of the buildings

- Preliminary Results & Discussion -

Effects of tasks and thinking styles on looking-around and task performance (Tab. 1, 2, Fig. 4)

Task types:

- People looked around more in the obj. col. task and less in the nav. task.
- In the learned env., new information is not necessary to navigate.

Interaction between task types and thinking styles:

- People with a stronger preference for effortful thinking (PET) or close-minded thinking looked around less during navigation task (Tab. 1, Fig. 4).
- Furthermore, higher PET participants completed the navigation task more quickly, while the effect of CMT was not observed (Tab. 2).

Interpretations and questions

- Participants with higher PET may learn better despite exhibiting similar amounts of look-around behavior as others and navigate more efficiently.
- CMT also affected the amount of LA during the navigation task, but not the performance. What was the difference in LA between CMT and PET during learning phase?

Tab. 1 Summary of GLME model for the total amount of LA during each trial. The model also included random effects of participants and buildings.

	<i>b</i>	<i>SE</i>		<i>b</i>	<i>SE</i>
(Interception)	4.948	.171	Age	.083	.049
Object collection	.831	.057	Obj. col. x Open-minded	-.081	.063
Navigation	-.860	.084	Nav. x Open-minded	.023	.096
Actively Open-minded	.003	.063	Obj. col. x Close-minded	.040	.058
Close-minded	.037	.057	Nav. x Close-minded	-.197	.088
Effortful	.045	.060	Obj. col. x Effortful	-.036	.061
Intuitive	-.043	.063	Nav. x Effortful	-.365	.086
SBSOD	.039	.054	Obj. col. x Intuitive	.004	.064
SBSOD ²	-.096	.033	Nav. x Intuitive	.001	.097

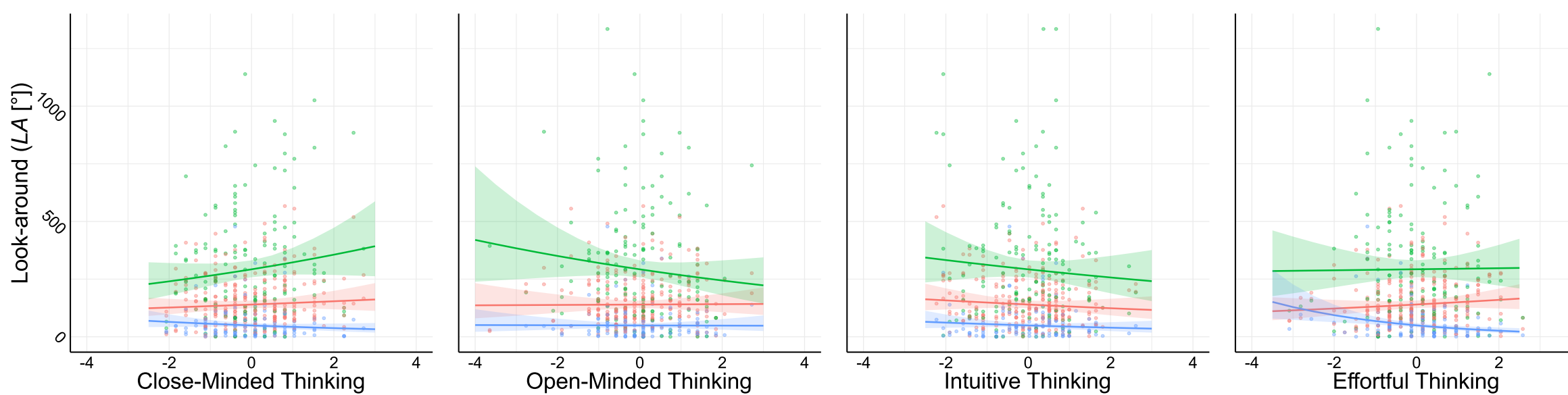


Fig. 4 Plots of relations between the total amount of LA during each trial and 4 components of thinking styles. Dots represent the observed value and lines with CI represent estimated values.

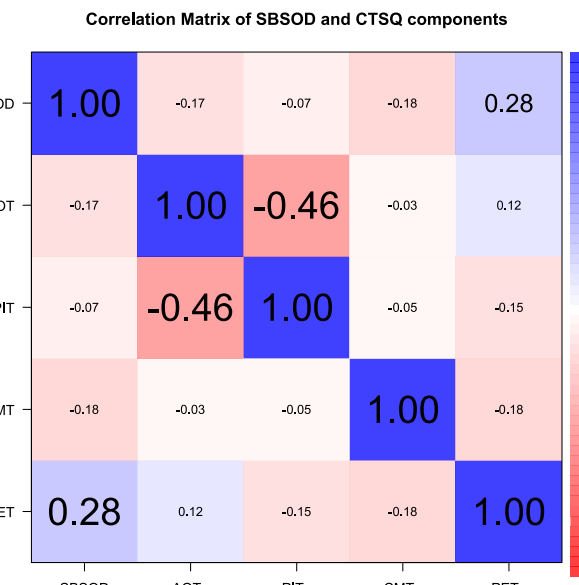


Fig. 5 Corr. matrix of SBSOD and 4 components of thinking styles.

Tab. 2 Summary of GLM model for the completion time of the navigation task.

	<i>b</i>	<i>SE</i>		<i>b</i>	<i>SE</i>
(Interception)	4.359	.127	SBSOD	.109	.073
Actively Open-minded	-.031	.079	Age	.105	.070
Close-minded	-.069	.065	Age ²	.082	.048
Effortful	-.139	.063	Intuitive ²	-.040	.049
Intuitive	-.061	.079	Close-minded ²	-.089	.051
Bldg. B	-.212	.130	SBSOD ²	.044	.041

- Ongoing Analysis & Future Direction -

More detailed analysis on spatiotemporal aspects

Spatial distribution of stopping and looking around

- Did participants with different thinking styles differ in where they stopped?

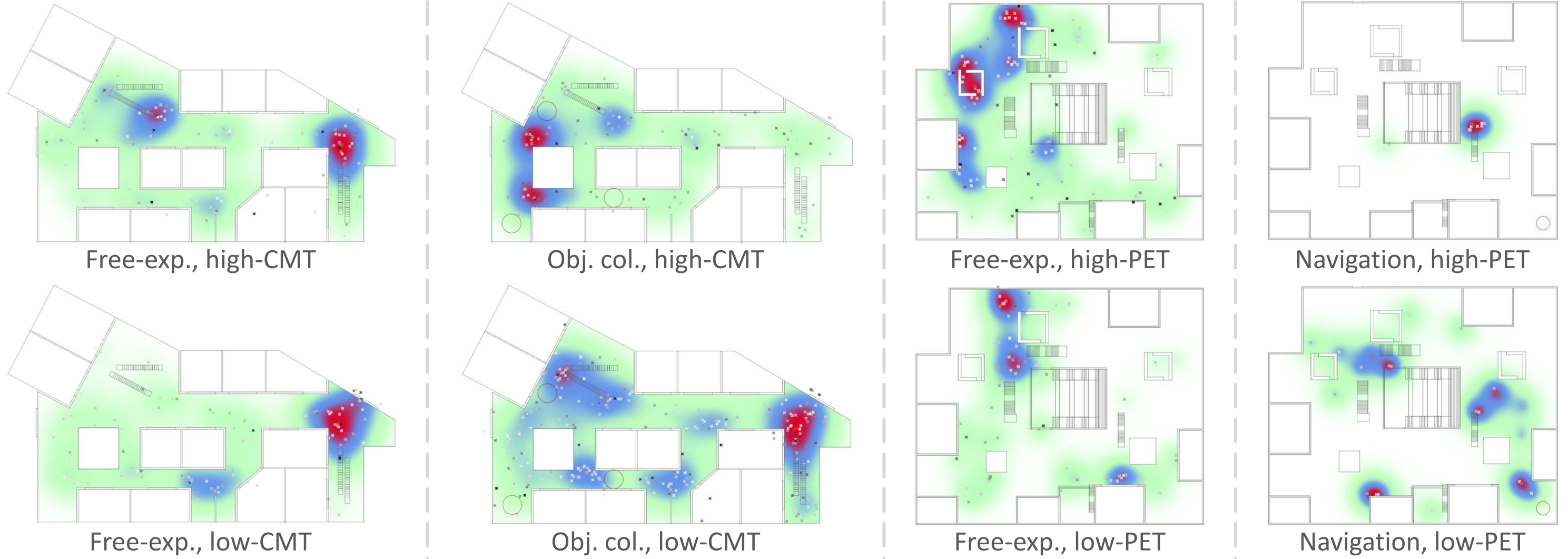


Fig. 6 Heatmaps showing where participants stopped during the tasks. Each map illustrates the stopping locations of participants with high or low PET/CMT scores. See additional maps via the QR code below.

Visuospatial analysis of looked around spaces based on ‘embodied 3D isovist’

- Did the spatial feature of viewed spaces differ despite similar amounts of LA?

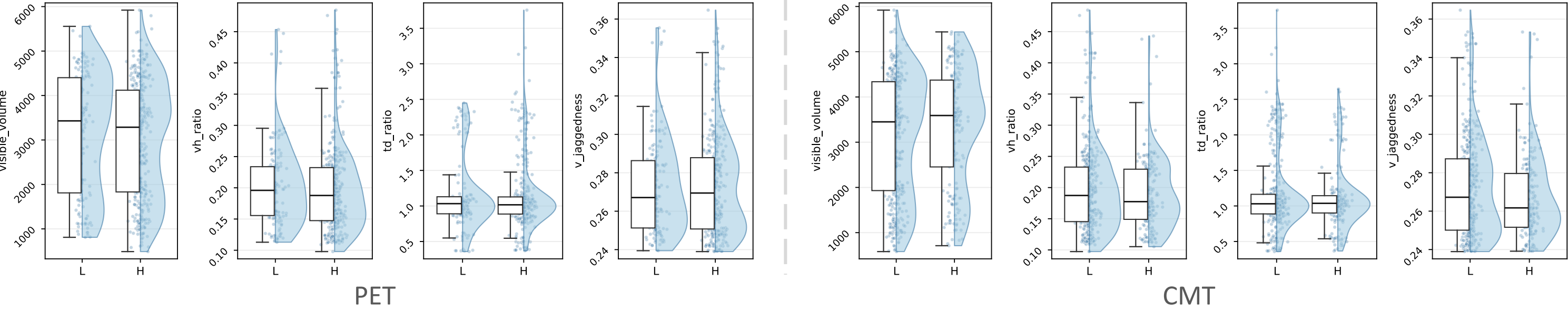


Fig. 7 Plots of embodied 3D isovist metrics of viewed spaces during stops. Results of the object collection task in Bldg. A are presented. See more from the QR code above.

- The embodied 3D isovist metrics of viewed space during stops and look-around did not show substantial differences between higher- and lower-PET (CMT) groups.

Detouring during the object collection task

- Did participants with higher PET make a detour after noticing the object and before collecting it?

Tab. 3 Summary of the GLM model for the directness of the route after noticing the object and before collecting it. Some ineffective variables are excluded due to the poster size (see more from the QR code).

	<i>b</i>	<i>SE</i>		<i>b</i>	<i>SE</i>
Actively Open-minded	.092	.039	Intuitive	.070	.038
Close-minded	-.027	.035	SBSOD	.079	.036
Effortful	-.044	.035	Length of detour ²	-.113	.039

- Contrary to the hypothesis, PET did not affect the detouring.
- Higher AOT and SBSOD were associated with less detouring.

Further exploration by using machine learning

- Possibility of developing machine learning models with high-dimensional data including the aforementioned factors.

- Acknowledgements -

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More info.

